

II-Probabilistic Approaches Empirical Distributions: Representation of Samples

Foundations of Neuro-Symbolic AI

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Observations of a Factored System

Example (Students and their semester)

We ask students about their semester and want to capture the statistics.

Task: Representation of Observations

Let there be observations of the system by a set of states

$$\{(x_0^j, \dots, x_{d-1}^j) : j \in [m]\}.$$

How to capture them by tensors?

Aim:

- ▶ Represent the observations for querying
- ▶ Find models making sense out of the observations

Formalization with a sample selector map

Given a dataset $\{(x_0^j, \dots, x_{d-1}^j) : j \in [m]\}$ of samples of the factored system we define the sample selector map

$$\beta^D : [m] \rightarrow \prod_{k \in [d]} [m_k]$$

Idea

Enrich the factored system with variables $X_0, \dots, X_{d-1}$ by a data selection variable L and encode β^D using the one-hot encoding of the larger system.

More general: Encoding of maps between Factored Systems

Definition

Let q be a function

$$q : \bigtimes_{k \in [d]} [m_k] \rightarrow \bigtimes_{l \in [r]} [m_l]$$

which maps the states of a factored system to variables $X_0, \dots, X_{d-1}$ to the states of another factored system with variables $Y_0, \dots, Y_{p-1}$. Then the tensor representation of q is a tensor

$$\beta^f \in \left(\bigotimes_{l \in [p]} \mathbb{R}^{m_l} \right) \otimes \left(\bigotimes_{k \in [d]} \mathbb{R}^{m_k} \right)$$

defined by

$$\beta^f = \sum_{x_0 \in [m_0]} \cdots \sum_{x_{k-1} \in [m_{k-1}]} \epsilon_{q(x_0, \dots, x_{d-1})} \otimes \epsilon_{x_0, \dots, x_{d-1}}.$$

Data tensor

In the case of the sample selection map

$$D : [m] \rightarrow \bigtimes_{k \in [d]} [m_k]$$

we have a representation tensor called the **data tensor**

$$\beta^D = \sum_{j \in [m]} \epsilon_j \otimes \epsilon_{x_0^j, \dots, x_{d-1}^j}$$

depicted as

Representation of Data

The one-hot encoding of each observation is stored on a slice of the data tensor.

Average of the Observation

The **empirical distribution** is the average of the one-hot encoded observed states:

$$\mathbb{P}^D := \frac{1}{m} \sum_{j \in [m]} \epsilon_{x_0^j, \dots, x_{d-1}^j}$$

In a contraction diagram we denote the average by

More efficient data representation

The storage demand of β^D increases exponentially

$$\dim [\beta^D] = \left(\prod_{k \in [d]} m_k \right) \cdot m .$$

Store the values X_k separately for each $k \in [d]$!

The collection of encoding tensors

$$\beta^{D_k} \in \mathbb{R}^{m_k \times m}$$

to the maps

$$D_k : [m] \rightarrow [m_k] \quad , \quad D_k(j) = x_k^j$$

has a linearly increasing storage demand of

$$\sum_{k \in [d]} \dim [\beta^{D_k}] = \sum_{k \in [d]} m_k \cdot m .$$

Representation of the Empirical in a CP Format

The data tensor can be stored by the contraction

$$\langle \beta^D \rangle_{[X_0, \dots, X_{k-1}, J]} = \langle \beta^{D_0}, \dots, \beta^{D_{d-1}} \rangle_{[X_0, \dots, X_{d-1}, J]}$$

depicted as

This is called a **CP Tensor Decomposition**.

CP Decomposition of Empirical Distributions

Instead of storing the datacore β^D , store the encodings β^{D_k} .

Log-likelihood

Likelihood of a dataset drawing data independently factorizes:

$$\prod_{j \in [m]} \mathbb{P} \left[x_{[d]}^j \right]$$

Taking the log and averaging:

This is the **cross-entropy** between the empirical distribution and the candidate distribution $\mathbb{P} \left[X_{[d]} \right]$.

Two learning tasks

- ▶ Structure learning: Find the hypergraph $\mathcal{G}$
- ▶ Parameter estimation: Find the best decorations by conditional probability distributions

Parameter estimation

Best log-likelihood maximizers are the conditional probability distributions.